

FOOD DELIVERY CUSTOMER ANALYSIS

Data Analyst Project Report

PostgreSQL • SQL • Power BI
Dataset: 388 customer records

1. Executive Summary

This project analyzes 388 online food-delivery customer records to understand customer segments and feedback patterns. The workflow covers data preparation, SQL-oriented analysis and an interactive Power BI dashboard.

2. Business Objective

- Understand customer distribution by key segments.
- Measure positive and negative customer feedback.
- Compare feedback across age, gender, occupation, income and education.
- Translate findings into practical business recommendations.

3. Dataset & Data Quality

- Records: 388
- Useful analytical columns: 13
- Positive feedback: 317 (81.70%)
- Negative feedback: 71 (18.30%)
- Output = Yes: 301 (77.58%)
- Missing values in analyzed fields: 0
- Unique full-row combinations: 285

4. Data Preparation

- Removed the unused unnamed CSV field from the analytical model.
- Trimmed categorical text values to avoid mismatches such as 'Negative ' vs 'Negative'.
- Checked missing values.
- Created Age Group for Power BI age analysis.
- Created KPI measures for Total Customers, Positive Feedback, Negative Feedback and Positive Feedback %.

5. Key Business Insights

- Overall satisfaction is high: 81.7% of customers provided positive feedback.
- Regular customers are the largest customer group with 218 records.
- Student has the highest observed positive feedback rate among occupations (89.9%).
- No Income has the highest observed positive feedback rate among income groups (90.9%).
- Post Graduate has the highest observed positive feedback rate among education groups (88.5%).

6. Power BI Dashboard

The final dashboard contains four KPI cards: Total Customers, Positive Feedback, Negative Feedback and Positive Feedback %.

It includes visuals for customer type, feedback by customer type, feedback by age group, gender, occupation, monthly income and education.

Slicers provide interactive filtering by Gender, Customer Type, Monthly Income and Age Group.

7. SQL Analysis Approach

- Use GROUP BY for customer-segment counts.
- Use conditional aggregation for positive/negative feedback rates.
- Use CTEs for multi-step business analysis.
- Use window functions when ranking segments is required.

8. Business Recommendations

- Prioritize retention and loyalty initiatives for regular customers.
- Investigate the operational reasons behind negative feedback.
- Personalize campaigns using segment-level insights.
- Track feedback KPIs regularly in Power BI.

9. Limitations

- The dataset contains 388 records, so findings are directional.
- There is no unique customer ID; repeated full rows cannot be confidently classified as duplicate customers.
- The analysis identifies associations rather than causal relationships.
- Operational metrics such as delivery time and order value are not included.

10. Conclusion

The dataset shows a 81.70% positive feedback rate. Segment-level analysis highlights differences across customer groups, while the Power BI dashboard communicates these findings interactively. The project demonstrates an end-to-end data analytics workflow suitable for a portfolio and interview discussion.